Package 'MRBEEX'

July 21, 2026

Type Package

Title Mendelian Randomization using Bias-Correction Estimating Equations with Extended Functions

Version 0.1.0

Author Yihe Yang

Maintainer Yihe Yang <yxy1234@case.edu>

Description

MRBEEX provides the MRBEE framework and several follow-up methods for Mendelian Randomization (MR) analysis. The package currently includes several methods, including CisMRBEEX, MRBEE_LDA, and MRBEE_TL: CisMRBEEX supports multivariable cis-Mendelian randomization, MRBEE_LDA supports LD-aware multivariable MR using bias-corrected estimating equations with optional SuSiE exposure selection, and MRBEE_TL supports transfer learning across populations. These methods are designed to reduce weak instrument bias, accommodate correlated instrumental variables through LD information, and improve causal effect estimation in settings with horizontal pleiotropy or population heterogeneity.

License MIT + file LICENSE

Encoding UTF-8

RoxygenNote 7.3.3

Imports MASS,
FDRestimation,
mvtnorm,
mixtools,
utils,
data.table,
varbvs,
susieR,
CppMatrix,
Matrix,
MRBEE

Remotes harryyiheyang/CppMatrix,
noahlorinczcomi/MRBEE,
stephenslab/susieR

ORCID 0000-0001-6563-3579

Contents

block_cutoff 2

CisMRBEEX . . . 3
CisMRBEE_UV . . . 6
cluster_snps_and_assign . . . 8
errorCov . . . 9
filter_align . . . 9
GWPT . . . 10
GWPT_Mixture . . . 11
mix_normal_generator . . . 12
MRBEEX_UV . . . 12
MRBEE_IMRP . . . 14
MRBEE_LDA . . . 15
MRBEE_TL . . . 19
MRBEE_TL_UV . . . 21
Sparse_Prediction . . . 23
summary_generation . . . 24

Index **26**

`block_cutoff` *Cluster pruning by Local Pratt Index (retain up to 5 per eligible cluster)*

Description

Given `cluster_info` produced by `clump_cluster` and two row-aligned effect-size vectors `marginal_effect` and `direct_effect`, this function prunes rows within clusters whose size exceeds `cutoff` by selecting, per such cluster, at most the top 5 variants ranked by the local Pratt index accumulating their normalized contributions until reaching `contribution`; if the threshold is not reached, the top 5 are kept. Clusters with size *less than or equal to* `cutoff` are fully retained.

Usage

```
block_cutoff(
  cluster_info,
  marginal_effect,
  direct_effect,
  cutoff = 5,
  max_size = 5,
  contribution = 0.9
)
```

Arguments

`cluster_info` A data frame from `clump_cluster`, containing at least a numeric column `cluster` giving the cluster index for each row; must be row-aligned with the effect vectors.

`marginal_effect` Numeric vector of GWAS marginal effect sizes, row-aligned with `cluster_info`. Must be standardized to a comparable scale (e.g., z-scores) before calling this function.

`direct_effect` Numeric vector of direct effect sizes (from fine-mapping or PRS), row-aligned with `cluster_info`. Must be standardized to a comparable scale before calling. If estimated by SBayesRC, a common standardization is $\beta * \sqrt{2 * f * (1 - f)}$, where f is the allele frequency.

`cutoff` Integer. Maximum cluster size threshold for pruning: clusters with size <= `cutoff` are entirely kept; clusters with size > `cutoff` are pruned using the Local Pratt rule.

`max_size` The maximum size of each cluster when using the Pratt index to cut off.

`contribution` Numeric in $(0, 1]$. Target cumulative contribution (based on normalized Local Pratt weights) used to decide how many top variants to retain per pruned cluster; no more than 5 will be kept.

Value

A list with:

- `cluster_info_updated`: the pruned subset of `cluster_info`
- `selected_global_idx`: global row indices retained
- `dropped_global_idx`: global row indices removed

`CisMRBEEX` *Cis Multivariable Mendelian Randomization using Bias-corrected Estimating Equations*

Description

This function performs multivariable cis-Mendelian randomization that removes weak instrument bias using Bias-corrected Estimating Equations and identifies uncorrelated horizontal pleiotropy (UHP). Additionally, it integrates SuSiE for exposure selection, enhancing interpretability.

Usage

```
CisMRBEEX(
  by,
  bX,
  byse,
  bXse,
  LD,
  Rxy,
  model.infinitesimal = F,
  reliability.thres = 0.6,
  Lvec = c(1:5),
  causal.pip.thres = 0.2,
  xQTL.selection.rule = "top_K",
  top_K = 1,
  xQTL.pip.min = 0.2,
  xQTL.max.L = 10,
  xQTL.cred.thres = 0.95,
  xQTL.pip.thres = 0.5,
  xQTL.Nvec,
```

```
  tauvec = seq(3, 30, by = 3),
  xQTL.weight = NULL,
  admm.rho = 2,
  ridge.diff = 1000,
  max.iter = 100,
  max.eps = 0.001,
  susie.iter = 500,
  coverage.xQTL = 0.95,
  coverage.causal = 0.95,
  ebic.theta = 0,
  ebic.gamma = 1,
  theta.ini = F,
  gamma.ini = F,
  xQTLfitList = NULL,
  standardize = F,
  verbose = T
)
```

Arguments

- `by` A vector of effect estimates from the outcome GWAS.
- `bX` A matrix of effect estimates from the exposure GWAS.
- `byse` A vector of standard errors of effect estimates from the outcome GWAS.
- `bXse` A matrix of standard errors of effect estimates from the exposure GWAS.
- `LD` The linkage disequilibrium (LD) matrix.
- `Rxy` The correlation matrix of estimation errors of exposures and outcome GWAS. The last column corresponds to the outcome.
- `model.infinitesimal` An indicator of whether using REML to model infinitesimal effects. Defaults to `F`.
- `reliability.thres` A threshold for the minimum value of the reliability ratio. If the original reliability ratio is less than this threshold, only part of the estimation error is removed so that the working reliability ratio equals this threshold. Default is `0.6`.
- `Lvec` When SuSiE is used, the candidate vector for the number of single effects. Default is `c(1:5)`.
- `causal.pip.thres` A threshold of minimum posterior inclusion probability. Default is `0.2`.
- `xQTL.selection.rule` The method for purifying informative xQTLs within each credible set. Options include "minimum_pip", which selects all variables with PIPs exceeding a specified threshold, and "top_K", which ensures at least K variables are selected based on their PIP ranking. Defaults to "top_K".
- `top_K` The maximum number of variables selected in each credible sets. Defaults to 1.
- `xQTL.pip.min` The minimum empirical PIP used in purifying variables in each credible set. Defaults to `0.2`.
- `xQTL.max.L` The maximum number of single effects used to estimate the xQTL effects. Defaults to 10.

`xQTL.cred.thres`
: The minimum empirical posterior inclusion probability (PIP) used in getting credible sets of xQTL selection. Defaults to `0.95`.

`xQTL.pip.thres` The threshold of individual PIP when selecting xQTL. Defaults to `0.5`.

`xQTL.Nvec` The vector of sample sizes of exposures.

`tauvec` The candidate vector of tuning parameters for the MCP penalty function. Default is `seq(3, 30, by=3)`.

`xQTL.weight` The vector of weights used in specifying the prior weights of SuSiE. Defaults to `NULL`.

`admm.rho` The tuning parameter in the nested ADMM algorithm. Default is `2`.

`ridge.diff` A ridge.parameter on the differences of causal effect estimate in one credible set. Defaults to `1e3`.

`max.iter` Maximum number of iterations for causal effect estimation. Defaults to `100`.

`max.eps` Tolerance for stopping criteria. Defaults to `0.001`.

`susie.iter` Number of iterations in SuSiE per iteration. Default is `500`.

`coverage.xQTL` The coverage of defining a credible set in xQTL selection. Defaults to `0.95`.

`coverage.causal`
: The coverage of defining a credible set in cis-MRBEE. Defaults to `0.95`.

`ebic.theta` EBIC factor on causal effect. Default is `0`.

`ebic.gamma` EBIC factor on horizontal pleiotropy Default is `1`.

`theta.ini` Initial value of theta. If `FALSE`, the default method is used to estimate. Default is `FALSE`.

`gamma.ini` Initial value of gamma. Default is `FALSE`.

`xQTLfitList` Initial SuSiE fits of xQTLs for exposures. This should be a list with one `susie.fit` object per exposure. Users can customize additional SuSiE parameters to improve performance. Default is `NULL`.

`standardize` If standardize = TRUE, standardize the columns of X to unit variance prior to fitting in SuSiE. Default is `F`.

`verbose` A logical indicator of whether to display the execution time of the method. Default is `T`.

Value

A list that contains the results of the MRBEEX with respect to different methods applied:

`theta` Causal effect estimate.

`theta.se` Standard error of the causal effect estimate.

`theta.cov` Covariance matrix of the causal effect estimate.

`theta.pip` Empirical posterior inclusion probability (PIP) of the causal effect in the subsampling procedure.

`theta.pratt` Pratt index estimate of exposure.

`susie.theta` The fit of causal effect resulted from susie.

`gamma` Estimate of horizontal pleiotropy.

`gamma.pratt` Pratt index estimate of horizontal pleiotropy.

`Bic` A vector or matrix recording the Bayesian Information Criterion (BIC) values.

`theta.ini` Initial value of `theta` used in the estimation procedure.

`gamma.ini` Initial value of `gamma` used in the estimation procedure.

`reliability.adjust` Estimated reliability-adjusted values.

`thetalist` List of `theta` estimates recorded during each iteration in the subsampling procedure.

`gammalist` List of `gamma` estimates recorded during each iteration in the subsampling procedure.

var.error The variance of residuals.

var.error The variance of infinitesimal effect.

estimated.reliability.ratio The estimated reliability ratios of exposures.

xQTLfitList The SuSiE sparse-prediction fits for the exposures.

`CisMRBEE_UV` *CisMRBEE_UV: Cis Univariable Mendelian Randomization using Bias-corrected Estimating Equations*

Description

This function performs univariable cis-Mendelian randomization that removes weak instrument bias using Bias-corrected Estimating Equations and identifies uncorrelated horizontal pleiotropy (UHP).

Usage

```
CisMRBEE_UV(
  by,
  bX,
  byse,
  bXse,
  LD,
  Rxy,
  xQTL.N,
  xQTL.selection.rule = "top_K",
  top_K = 1,
  xQTL.pip.min = 0.2,
  xQTL.max.L = 10,
  xQTL.cred.thres = 0.95,
  xQTL.pip.thres = 0.5,
  reliability.thres = 0.6,
  tauvec = seq(3, 30, by = 1.5),
  admm.rho = 2,
  coverage.xQTL = 0.95,
  coverage.causal = 0.95,
  max.iter = 100,
  max.eps = 0.001,
  ebic.gamma = 1,
  xQTLfit = NULL
)
```

Arguments

`by`	A vector of effect estimates from the outcome GWAS.
`bX`	A vector of effect estimates from the exposure GWAS.
`byse`	A vector of standard errors of effect estimates from the outcome GWAS.
`bXse`	A vector of standard errors of effect estimates from the exposure GWAS.
`LD`	The LD matrix of variants.
`Rxy`	The correlation matrix of estimation errors of exposures and outcome GWAS. The last column corresponds to the outcome.
`xQTL.N`	The sample sizes of exposure.
`xQTL.selection.rule`	The method for purifying informative xQTLs within each credible set. Options include "minimum_pip", which selects all variables with PIPs exceeding a specified threshold, and "top_K", which ensures at least K variables are selected based on their PIP ranking. Defaults to "top_K".
`top_K`	The maximum number of variables selected in each credible sets. Defaults to 1.
`xQTL.pip.min`	The minimum empirical PIP used in purifying variables in each credible set. Defaults to `0.2`.
`xQTL.max.L`	The maximum number of L in estimating the xQTL effects. Defaults to 10.
`xQTL.cred.thres`	The minimum empirical posterior inclusion probability (PIP) used in getting credible sets of xQTL selection. Defaults to `0.95`.
`xQTL.pip.thres`	If SuSiE fails to find any credible set, the threshold of individual PIP when selecting xQTL. Defaults to `0.5`.
`reliability.thres`	A threshold for the minimum value of the reliability ratio. If the original reliability ratio is less than this threshold, only part of the estimation error is removed so that the working reliability ratio equals this threshold. Default is `0.6`.
`tauvec`	A vector of tuning parameters used in penalizing the direct causal effect. Default is `seq(3, 30, by=1.5)`.
`admm.rho`	A parameter set in the ADMM algorithm. Default is `2`.
`coverage.xQTL`	The coverage of defining a credible set in xQTL selection. Defaults to `0.95`.
`coverage.causal`	The coverage of defining a credible set in cis-MRBEE. Defaults to `0.95`.
`max.iter`	The maximum number of iterations for the ADMM algorithm. Default is `100`.
`max.eps`	The convergence tolerance for the ADMM algorithm. Default is `0.001`.
`ebic.gamma`	The extended BIC factor for model selection. Default is `1`.
`xQTLfit`	Initial SuSiE fit of the xQTL effects. Default is `NULL`.

Value

A list containing:

`theta` The estimated effect size of the tissue-gene pair.

`gamma` The estimated effect sizes of the direct causal variants.

`theta.cov` The variance of the estimated effect size ‘theta‘.

`theta.se` The standard error of the estimated effect size 'theta'.

`theta.z` The z-score of the estimated effect size 'theta'.

`Bic` The BIC values for each tuning parameter.

`eQTL.fit` The SuSiE result of xQLT selection of exposure.

`var.error` The variance of residuals.

`var.error` The variance of infinitesimal effect.

`causal.fit` The SuSiE result of causal effect calibration of exposure.

`reliability.adjust` Estimated reliability-adjusted values.

`cluster_snps_and_assign` *Cluster SNPs (greedy by P) and assign every SNP to exactly one cluster*

Description

One-pass pipeline for C+T style clustering and membership assignment. For each chromosome, repeatedly take the SNP with the smallest P as a center, group all SNPs within `±window_size` bp into that cluster, remove them, and continue until all SNPs are clustered. Then assign every SNP in `df` to exactly one cluster by non-equi interval join with explicit tie-breaking in overlapping regions: smallest distance to center, then smaller center P, then smaller `cluster_id`.

Usage

```
cluster_snps_and_assign(df, window_size = 1e+06)
```

Arguments

`df`	A data.frame/data.table with columns: `SNP`, `CHR`, `BP`, and `P`.
`window_size`	Integer, window radius in base pairs (e.g., `1e6` for ±1 Mb).

Value

A list with two data.tables:

- `centers`: cluster centers (like df1), columns `SNP`, `CHR`, `BP`, `P`, `ClusterSize`, `left`, `right`, `cluster_id`.
- `assigned`: original `df` annotated with `cluster` (ID), distance to center `dist_bp`, and center metadata `center_SNP`, `center_BP`, `center_P`.

`errorCov` *Estimate Error Covariance Matrix Using GWAS Insignificant Effects*

Description

This function estimates the error covariance matrix by subsampling a proportion of insignificant GWAS effects and calculating their correlation coefficients.

Usage

```
errorCov(
  ZMatrix,
  Zscore.cutoff = 2,
  subsampling.ratio = 0.1,
  subsampling.time = 1000
)
```

Arguments

`ZMatrix`	A matrix of Z-scores for exposure and outcome, with the outcome GWAS in the last column.
`Zscore.cutoff`	The cutoff for significance. Defaults to 2.
`subsampling.ratio`	The proportion of effects to subsample for each iteration. Defaults to 0.1.
`subsampling.time`	The number of subsampling iterations. Defaults to 1000.

Value

A matrix representing the estimated error covariance.

`filter_align` *Filter and Align GWAS Data to a Reference Panel*

Description

The `filter_align` function processes a list of GWAS summary statistics data frames, harmonizes alleles according to a reference panel, removes duplicates, and aligns data to common SNPs.

Usage

```
filter_align(gwas_data_list, ref_panel, allele_match = T)
```

Arguments

`gwas_data_list`	A list of data.frames where each data.frame contains GWAS summary statistics for a trait. Each data.frame should include columns for SNP identifiers, Z-scores of effect size estimates, sample sizes (N), effect allele (A1), and reference allele (A2).
`ref_panel`	A data.frame containing the reference panel data. It must include columns for SNP, A1, and A2.
`allele_match`	An indicator of whether matching the effect alleles of GWAS files to the reference panel.

Details

The function performs several key steps: adjusting alleles according to a reference panel, removing duplicate SNPs, and aligning all GWAS data frames to a set of common SNPs. This is often a necessary preprocessing step before performing genetic correlation and heritability analyses.

Value

A list of data.frames, each corresponding to an input GWAS summary statistics data frame, but filtered, harmonized, and aligned to the common SNPs found across all data frames.

`GWPT` *Genome-Wide Pleiotropy Test*

Description

This function performs a genome-wide pleiotropy test (GWPT) after Mendelian randomization. It offers an option for a two-mixture model, where the residual is chosen as the smaller one resulting from the two causal effect estimates from two mixtures.

Usage

```
GWPT(by, byse, bX, bXse, Rxy, theta, theta.cov)
```

Arguments

`by`	A vector of effect estimates from the outcome GWAS.
`byse`	A vector of standard errors of effect estimates from the outcome GWAS.
`bX`	A matrix of effect estimates from the exposure GWAS.
`bXse`	A matrix of standard errors of effect estimates from the exposure GWAS.
`Rxy`	The correlation matrix of estimation errors of exposures and outcome GWAS. The last column corresponds to the outcome.
`theta`	The causal effect estimate.
`theta.cov`	The covariance matrix of the causal effect estimate.

Value

A list with two components:

`BETA`	The estimated residual values.
`SE`	The standard errors of the residual estimates.

GWPT_Mixture *Genome-Wide Pleiotropy Test for mixture model*

Description

This function performs a genome-wide pleiotropy test (GWPT) after Mendelian randomization. It offers an option for a two-mixture model, where the residual is chosen as the smaller one resulting from the two causal effect estimates from two mixtures.

Usage

```
GWPT_Mixture(
  by,
  byse,
  bX,
  bXse,
  Rxy,
  theta1,
  theta.cov1,
  theta2,
  theta.cov2,
  LD.block
)
```

Arguments

by	A vector of effect estimates from the outcome GWAS.
byse	A vector of standard errors of effect estimates from the outcome GWAS.
bX	A matrix of effect estimates from the exposure GWAS.
bXse	A matrix of standard errors of effect estimates from the exposure GWAS.
Rxy	The correlation matrix of estimation errors of exposures and outcome GWAS. The last column corresponds to the outcome.
theta1	The causal effect estimate of the first mixture.
theta.cov1	The covariance matrix of the causal effect estimate of the first mixture.
theta2	The causal effect estimate of the second mixture.
theta.cov2	The covariance matrix of the causal effect estimate of the second mixture.
LD.block	A vector of indices of LD blocks.

Value

A list with two components:

BETA	The estimated residual values.
SE	The standard errors of the residual estimates.

`mix_normal_generator` *Generate data from a Mixture of Normal Distributions (Variance Contribution Model)*

Description

This function generates samples from a mixture of zero-mean normal distributions. The 'var' parameter specifies the total variance contribution of each component (i.e., pi_i * sigma_i^2), not the component variance itself.

Usage

```
mix_normal_generator(
  m,
  prop = c(0.9, 0.09, 0.009, 9e-04),
  var = c(0.35, 0.4, 0.2, 0.05)
)
```

Arguments

`m`	The total number of random samples to generate.
`prop`	A numeric vector of mixing proportions (weights). These are normalized. Default: c(0.9, 0.09, 0.009, 0.0009).
`var`	A numeric vector where var[i] represents the variance contribution of component i: V_i = prop[i] * sigma_i^2. Default: c(0.35, 0.4, 0.2, 0.05).

Value

A numeric vector of length 'm' containing the random variates.

`MRBEEX_UV` *Univariable Mendelian Randomization using Bias-corrected Estimating Equations*

Description

This function removes weak instrument bias using Bias-corrected Estimating Equations and identifies uncorrelated horizontal pleiotropy (UHP) and correlated horizontal pleiotropy (CHP) through IPOD or greedy search.

Usage

```
MRBEEX_UV(
  by,
  bX,
  byse,
  bXse,
  LD = "identity",
  Rxy,
```

```
  cluster.index = c(1:length(by)),
  reliability.thres = 0.6,
  Method = "IPOD",
  tauvec = seq(4, 8, by = 2),
  admm.rho = 2,
  ebic.gamma = 1,
  Kvec = seq(1, length(bX)/2, by = 2),
  max.iter = 100,
  max.eps = 0.001,
  maxdiff = 3,
  sampling.time = 1000,
  sampling.iter = 20,
  theta.ini = F,
  gamma.ini = F
)
```

Arguments

`by`	A vector of effect estimates from the outcome GWAS.
`bX`	A vector of effect estimates from the exposure GWAS.
`byse`	A vector of standard errors of effect estimates from the outcome GWAS.
`bXse`	A vector of standard errors of effect estimates from the exposure GWAS.
`LD`	The linkage disequilibrium (LD) matrix. Default is the identity matrix, assuming independent instrumental variables (IVs).
`Rxy`	The correlation matrix of estimation errors of exposures and outcome GWAS. The last element corresponds to the outcome.
`cluster.index`	A vector indicating the LD block indices each IV belongs to. The length is equal to the number of IVs, and values are the LD block indices.
`reliability.thres`	A threshold for the minimum value of the reliability ratio. If the original reliability ratio is less than this threshold, only part of the estimation error is removed so that the working reliability ratio equals this threshold. Default is `0.6`.
`Method`	Method for handling horizontal pleiotropy. Options are `"IPOD"` and `"Greedy"`.
`tauvec`	When choosing `"IPOD"`, the candidate vector of tuning parameters for the MCP penalty function. Default is `seq(4, 8, by=2)`.
`admm.rho`	When choosing `"IPOD"`, the tuning parameter in the nested ADMM algorithm. Default is `2`.
`ebic.gamma`	EBIC factor on horizontal pleiotropy Default is `1`.
`Kvec`	When choosing `"Greedy"`, the candidate vector of number of pleiotropy in the greedy search algorithm. Default is `seq(1, length(bX)/2, by=2)`.
`max.iter`	Maximum number of iterations for causal effect estimation. Defaults to `100`.
`max.eps`	Tolerance for stopping criteria. Defaults to `0.001`.
`maxdiff`	The maximum difference between the MRBEE causal estimate and the initial estimator. Defaults to `3`.
`sampling.time`	Number of resampling times. Default is `1000`.
`sampling.iter`	Number of iterations per resampling. Default is `20`.
`theta.ini`	Initial value of theta. If `FALSE`, the default method is used to estimate. Default is `FALSE`.
`gamma.ini`	Initial value of gamma. Default is `FALSE`.

Value

A list containing the results of the MRBEEX.UV analysis:

`theta` Causal effect estimate.

`theta.se` Standard error of the causal effect estimate.

`theta.cov` Covariance matrix of the causal effect estimate.

`theta.bootstrap` Resampled causal effect estimates.

`gamma` Estimate of horizontal pleiotropy.

`Bic` A vector or matrix recording the Bayesian Information Criterion (BIC) values.

`theta.ini` Initial value of `theta` used in the estimation procedure.

`gamma.ini` Initial value of `gamma` used in the estimation procedure.

`reliability.adjust` Estimated reliability-adjusted values.

`thetalist` List of `theta` estimates recorded during each iteration in the resampling procedure.

`gammalist` List of `gamma` estimates recorded during each iteration in the resampling procedure.

MRBEE_IMRP — *Mendelian randomization with bias-correction estimating equation: detecting horizontal pleiotropy via hypothesis test.*

Description

This function estimates the causal effect using a bias-correction estimating equation, considering potential pleiotropy and measurement errors.

Usage

```
MRBEE_IMRP(
  by,
  bX,
  byse,
  bXse,
  Rxy,
  max.iter = 30,
  max.eps = 1e-04,
  pv.thres = 0.05,
  var.est = "variance",
  FDR = T,
  adjust.method = "Sidak",
  maxdiff = 1.5,
  group.penalize = F,
  group.index = NULL,
  group.diff = 1000
)
```

Arguments

by	A vector (n x 1) of the GWAS effect size of outcome.
bX	A matrix (n x p) of the GWAS effect sizes of p exposures.
byse	A vector (n x 1) of the GWAS effect size SE of outcome.
bXse	A matrix (n x p) of the GWAS effect size SEs of p exposures.
Rxy	A matrix (p+1 x p+1) of the correlation matrix of the p exposures and outcome. The last one should be the outcome.
max.iter	Maximum number of iterations for causal effect estimation. Defaults to 30.
max.eps	Tolerance for stopping criteria. Defaults to 1e-4.
pv.thres	P-value threshold in pleiotropy detection. Defaults to 0.05.
var.est	Method for estimating the variance of residual in pleiotropy test. Can be "robust", "variance", or "ordinal". Defaults is "variance" that estimates the variance of residual using median absolute deviation (MAD).
FDR	Logical. Whether to apply the FDR to convert the p-value to q-value. Defaults to TRUE.
adjust.method	Method for estimating q-value. Defaults to "Sidak".
maxdiff	The maximum difference between the MRBEE causal estimate and the initial estimator. Defaults to 1.5.
group.penalize	An indicator of whether using SuSiE to penalize highly correlated exposures. Defaults to `F`.
group.index	A vector of the group index of exposure. Defaults to `c(1:ncol(bX))`.
group.diff	The tuning penalizing difference of highly correlated exposure prediction. Defaults to `1000`.

Value

A list containing the estimated causal effect, its covariance, and pleiotropy

---

MRBEE_LDA *MRBEE_LDA for LD-aware Multivariable Mendelian Randomization*

Description

MRBEE_LDA is the LD-aware multivariable MR component in the MRBEEX package. It removes weak instrument bias using bias-corrected estimating equations and identifies uncorrelated horizontal pleiotropy (UHP) using the IPOD algorithm, where outliers are interpreted as UHP. When `use.susie = FALSE`, the function returns the direct IPOD result without resampling. When `use.susie = TRUE`, it integrates SuSiE for exposure selection and uses the newer fixed half-SNP sampling procedure for inference.

Usage

```
MRBEE_LDA(
  by,
  bX,
  byse,
  bXse,
  LD = "identity",
  Rxy,
  cluster.index = c(1:length(by)),
  use.susie = T,
  inference = "subsampling",
  epsilon = 0.5,
  standardize = F,
  group.penalize = F,
  group.index = c(1:ncol(bX)[1]),
  group.diff = 100,
  tauvec = seq(4, 8, by = 0.5),
  admm.rho = 2,
  Lvec = c(1:min(10, ncol(bX))),
  pip.thres = 0.25,
  estimate_residual_variance = T,
  pip.min = 0.1,
  cred.pip.thres = 0.95,
  max.iter = 50,
  max.eps = 1e-05,
  susie.iter = 100,
  ebic.theta = 0,
  ebic.gamma = 1,
  ridge.diff = 1000,
  estimate_residual_method = "MoM",
  sampling.time = 300,
  sampling.iter = 25,
  group.size = 4,
  maxdiff = 3,
  reliability.thres = 0.6,
  coverage.causal = 0.95,
  theta.ini = F,
  gamma.ini = F,
  verbose = T,
  projection.eigen.floor = 1
)
```

Arguments

by	A vector of effect estimates from the outcome GWAS.
bX	A matrix of effect estimates from the exposure GWAS.
byse	A vector of standard errors of effect estimates from the outcome GWAS.
bXse	A matrix of standard errors of effect estimates from the exposure GWAS.
LD	The linkage disequilibrium (LD) matrix. Default is the identity matrix, assuming independent instrumental variables (IVs).

- `Rxy` The correlation matrix of estimation errors of exposures and outcome GWAS. The last column corresponds to the outcome.
- `cluster.index` A vector indicating the LD block indices each IV belongs to. The length is equal to the number of IVs, and values are the LD block indices.
- `use.susie` An indicator of whether using SuSiE to select causal exposures. Defaults to `T`.
- `inference` Inference method when `use.susie = TRUE`. `"subsampling"` (default) uses grouped LD subsampling; `"datafission"` uses one Gaussian selection view and an independent fixed-support oracle inference view.
- `epsilon` Information fraction allocated to the data-fission selection view. Must be between `0.1` and `0.9`, inclusive. Default is `0.5`.
- `standardize` If standardize = TRUE, standardize the columns of X to unit variance prior to fitting (or equivalently standardize XtX and Xty to have the same effect) in SuSiE. Note that scaled_prior_variance specifies the prior on the coefficients of X after standardization (if it is performed). If you do not standardize, you may need to think more carefully about specifying scaled_prior_variance. Whatever your choice, the coefficients returned by coef are given for X on the original input scale. Any column of X that has zero variance is not standardized.
- `group.penalize` An indicator of whether using difference penalty to penalize highly correlated exposures. Defaults to `F`.
- `group.index` A vector of the group index of exposure. Defaults to `c(1:ncol(bX))`.
- `group.diff` The tuning penalizing difference of highly correlated exposure prediction. Defaults to `100`.
- `tauvec` The candidate vector of tuning parameters for the MCP penalty function. Default is `seq(4, 8, by=0.5)`.
- `admm.rho` The tuning parameter in the nested ADMM algorithm. Default is `2`.
- `Lvec` When SuSiE is used, the candidate vector for the number of single effects. Default is `c(1:min(10, ncol(bX)))`.
- `pip.thres` Posterior inclusion probability (PIP) threshold. Individual PIPs less than this value will be shrunk to zero. Default is `0.25`.
- `estimate_residual_variance` When SuSiE is used, an indicator of whether estimating the variance of residuals. If setting F, the variance of residual will be fixed as `Rxy[p+1,p+1]`. Default is `T`.
- `pip.min` The minimum empirical PIP used in purifying variables in each credible set. Defaults to `0.1`.
- `cred.pip.thres` The threshold of PIP of each credible set. Defaults to `0.95`.
- `max.iter` Maximum number of iterations for causal effect estimation. Defaults to `50`.
- `max.eps` Tolerance for stopping criteria. Defaults to `1e-5`.
- `susie.iter` Number of iterations in SuSiE per iteration. Default is `100`.
- `ebic.theta` EBIC factor on causal effect. Default is `0`.
- `ebic.gamma` EBIC factor on horizontal pleiotropy. Default is `1`.
- `ridge.diff` A ridge.parameter on the differences of causal effect estimate in one credible set. Defaults to `1e3`.
- `estimate_residual_method` The method used for estimating residual variance. For the original SuSiE model, "MLE" and "MoM" estimation is equivalent, but for the infinitesimal model, "MoM" is more stable.

`sampling.time` Number of fixed half-SNP sampling repeats when `use.susie = TRUE`. This is not used when `use.susie = FALSE`. Default is `300`.

`sampling.iter` Number of estimation iterations per sampling repeat when `use.susie = TRUE`. This is not used when `use.susie = FALSE`. Default is `25`.

`group.size` Target size for block-aware subsampling groups when `use.susie = TRUE`. Singleton LD blocks are sampled independently with probability 0.5, and size-two LD blocks are sampled intact. Larger blocks are randomly reordered between blocks while preserving their internal IV order, concatenated, divided into groups of this size, and half of the groups are sampled. Default is `4`.

`maxdiff` The maximum difference between the MRBEE causal estimate and the initial estimator. Defaults to `3`.

`reliability.thres` A threshold for the minimum value of the reliability ratio. If the original reliability ratio is less than this threshold, only part of the estimation error is removed so that the working reliability ratio equals this threshold. Default is 0.6.

`coverage.causal` The coverage of defining a credible set when `use.susie = T`. Defaults to `0.95`.

`theta.ini` Initial value of theta. If `FALSE`, the default method is used to estimate. Default is `FALSE`.

`gamma.ini` Initial value of gamma. Default is `FALSE`.

`verbose` A logical indicator of whether to display the execution time of the method. Default is `T`.

`projection.eigen.floor` The minimum eigenvalue used when projecting SuSiE and selected refit cross-product matrices. The full-data floor is this value; resampled matrices are scaled by their current row count divided by the full row count. Defaults to `1`.

Value

A list that contains the results of MRBEE_LDA:

`theta` Causal effect estimate.

`theta.se` Standard error of the causal effect estimate.

`theta.cov` Covariance matrix of the causal effect estimate.

`theta.pip` Empirical posterior inclusion probability (PIP) of the causal effect when `use.susie = TRUE`.

`gamma` Estimate of horizontal pleiotropy.

`Bic` A vector or matrix recording the Bayesian Information Criterion (BIC) values.

`theta.ini` Initial value of `theta` used in the estimation procedure.

`gamma.ini` Initial value of `gamma` used in the estimation procedure.

`reliability.adjust` A global vector of covariance scaling factors.

`thetalist` List of `theta` estimates recorded during the sampling procedure when available.

`gammalist` List of `gamma` estimates recorded during the sampling procedure when available.

MRBEE_TL *Estimate Non-Transferable Causal Effect with MRBEE and SuSiE*

Description

This function estimates the non-transferable causal effect using a bias-correction estimating equation, considering potential pleiotropy and measurement errors, and using SuSiE to select the non-transferable causal effect.

Usage

```
MRBEE_TL(
  by,
  bX,
  byse,
  bXse,
  Rxy,
  LD = "identity",
  cluster.index = c(1:length(by)),
  group.penalize = F,
  group.index = NULL,
  group.diff = 100,
  theta.source,
  theta.source.cov,
  tauvec = seq(4, 8, 0.5),
  Lvec = c(1:6),
  standardize = F,
  admm.rho = 2,
  ebic.delta = 0,
  ebic.gamma = 1,
  transfer.coef = 1,
  susie.iter = 200,
  pip.thres = 0.25,
  pip.min = 0.1,
  cred.pip.thres = 0.95,
  max.iter = 50,
  coverage.causal = 0.95,
  max.eps = 1e-06,
  reliability.thres = 0.6,
  ridge.diff = 100,
  estimate_residual_method = "MoM",
  sampling.strategy = "bootstrap",
  group.size = 4,
  projection.eigen.floor = 1,
  sampling.time = 300,
  sampling.iter = 25
)
```

Arguments

by	A vector (n x 1) of the GWAS effect size of outcome.

bX	A matrix (n x p) of the GWAS effect sizes of p exposures.
byse	A vector (n x 1) of the GWAS effect size SE of outcome.
bXse	A matrix (n x p) of the GWAS effect size SEs of p exposures.
Rxy	A matrix (p+1 x p+1) of the correlation matrix of the p exposures and outcome. The first one should be the transferred linear predictor and last one should be the outcome.
LD	The linkage disequilibrium (LD) matrix. Default is the identity matrix, assuming independent instrumental variables (IVs).
cluster.index	A vector indicating the LD block indices each IV belongs to. The length is equal to the number of IVs, and values are the LD block indices.
group.penalize	An indicator of whether using difference penalty to penalize highly correlated exposures. Defaults to F.
group.index	A vector of the group index of exposure. Defaults to NULL.
group.diff	The tuning penalizing difference of highly correlated exposure prediction. Defaults to 100.
theta.source	A vector (p x 1) of the causal effect estimate learning from the source data.
theta.source.cov	A matrix (p x p) of the covariance matrix of the causal effect estimate learning from the source data.
tauvec	The candidate vector of tuning parameters for the MCP penalty function. Default is seq(4, 8, by=0.5).
Lvec	A vector of the number of single effects used in SuSiE. Default is c(1:6).
standardize	If standardize = TRUE, standardize the columns of X to unit variance prior to fitting (or equivalently standardize XtX and Xty to have the same effect) in SuSiE. Note that scaled_prior_variance specifies the prior on the coefficients of X after standardization (if it is performed). If you do not standardize, you may need to think more carefully about specifying scaled_prior_variance. Whatever your choice, the coefficients returned by coef are given for X on the original input scale. Any column of X that has zero variance is not standardized.
admm.rho	When choosing "IPOD", the tuning parameter in the nested ADMM algorithm. Default is 2.
ebic.delta	A scale of tuning parameter of causal effect estimate in extended BIC. Default is 0.
ebic.gamma	A scale of tuning parameter of horizontal pleiotropy in extended BIC. Default is 1.
transfer.coef	A scale of transfer.coef of theta.source to theta.target. Default is 1.
susie.iter	A scale of the maximum number of iterations used in SuSiE. Default is 200.
pip.thres	Posterior inclusion probability (PIP) threshold. Individual PIPs less than this value will be shrunk to zero. Default is 0.25.
pip.min	The minimum empirical PIP used in purifying variables in each credible set. Defaults to 0.1.
cred.pip.thres	The threshold of PIP of each credible set. Defaults to 0.95.
max.iter	Maximum number of iterations for causal effect estimation. Default is 50.
coverage.causal	The coverage of defining a credible set in MRBEEX when use.susie = T. Defaults to 0.95.

`max.eps` Tolerance for stopping criteria. Default is `1e-6`.

`reliability.thres`
: A scale of threshold for the minimum value of the reliability ratio. If the original reliability ratio is less than this threshold, only part of the estimation error is removed so that the working reliability ratio equals this threshold. Default is `0.6`.

`ridge.diff` A scale of parameter on the differences of causal effect estimate in one credible set. Defaults to `100`.

`estimate_residual_method`
: The method used for estimating residual variance. For the original SuSiE model, "MLE" and "MoM" estimation is equivalent, but for the infinitesimal model, "MoM" is more stable.

`sampling.strategy`
: Resampling scheme used by the independent branch where `LD="identity"`. For LD-aware analyses, only `"subsampling"` is allowed.

`group.size` Number of adjacent LD-ordered IVs per subsampling group in the LD-aware branch. `group.size=1` gives the original IV-level subsampling. Default is `4`.

`projection.eigen.floor`
: The minimum eigenvalue used when projecting SuSiE and selected refit cross-product matrices. The full-data floor is this value; resampled matrices are scaled by their current row count divided by the full row count. Defaults to `1`.

`sampling.time` Number of fixed half-SNP sampling repeats for standard-error estimation. Default is `300`.

`sampling.iter` Number of estimation iterations per sampling repeat. Default is `25`.

Value

A list containing the estimated causal effect, its covariance, and pleiotropy.

`MRBEE_TL_UV` *Estimate Non-Transferable Causal Effect with MRBEE and SuSiE in UVMR*

Description

This function estimates the non-transferable causal effect using a bias-correction estimating equation in a univariable MR model, considering potential pleiotropy and measurement errors, and using SuSiE to select the non-transferable causal effect.

Usage

```
MRBEE_TL_UV(
  by,
  bX,
  byse,
  bXse,
  Rxy,
  LD = "identity",
  cluster.index = c(1:length(by)),
  theta.source,
```

```
  theta.source.cov,
  tauvec = seq(3, 30, 3),
  admm.rho = 2,
  ebic.delta = 1,
  ebic.gamma = 1,
  transfer.coef = 1,
  susie.iter = 200,
  pip.thres = 0.3,
  max.iter = 50,
  max.eps = 1e-04,
  reliability.thres = 0.6,
  sampling.time = 100,
  sampling.iter = 10
)
```

Arguments

by	A vector (n x 1) of the GWAS effect size of outcome.
bX	A matrix (n x p) of the GWAS effect sizes of p exposures.
byse	A vector (n x 1) of the GWAS effect size SE of outcome.
bXse	A matrix (n x p) of the GWAS effect size SEs of p exposures.
Rxy	A matrix (p+1 x p+1) of the correlation matrix of the p exposures and outcome. The first one should be the transferred linear predictor and last one should be the outcome.
LD	The linkage disequilibrium (LD) matrix. Default is the identity matrix, assuming independent instrumental variables (IVs).
cluster.index	A vector indicating the LD block indices each IV belongs to. The length is equal to the number of IVs, and values are the LD block indices.
theta.source	A vector (p x 1) of the causal effect estimate learning from the source data.
theta.source.cov	A matrix (p x p) of the covariance matrix of the causal effect estimate learning from the source data.
tauvec	The candidate vector of tuning parameters for the MCP penalty function. Default is `seq(3, 30, by=3)`.
admm.rho	When choosing `"IPOD"`, the tuning parameter in the nested ADMM algorithm. Default is `2`.
ebic.delta	A scale of tuning parameter of causal effect estimate in extended BIC. Default is `1`.
ebic.gamma	A scale of tuning parameter of horizontal pleiotropy in extended BIC. Default is `1`.
transfer.coef	A scale of transfer.coef of theta.source to theta.target. Default is `1`.
susie.iter	A scale of the maximum number of iterations used in SuSiE. Default is `200`.
pip.thres	A scale of PIP theshold for calibyating causality used in SuSiE. Default is `0.3`.
max.iter	Maximum number of iterations for causal effect estimation. Default is `50`.
max.eps	Tolerance for stopping criteria. Default is `1e-4`.

reliability.thres
: A scale of threshold for the minimum value of the reliability ratio. If the original reliability ratio is less than this threshold, only part of the estimation error is removed so that the working reliability ratio equals this threshold. Default is `0.6`.

sampling.time
: Number of resampling repeats for standard-error estimation. Default is `100`.

sampling.iter
: Number of estimation iterations per resampling repeat. Default is `10`.

Value

A list containing the estimated causal effect, its covariance, and pleiotropy.

`Sparse_Prediction` *Sparse Prediction of xQTL Effects using SuSiE*

Description

This function performs informative xQTL selection and sparse prediction using SuSiE.

Usage

```
Sparse_Prediction(
  bX,
  bXse,
  LD,
  xQTL.Nvec,
  xQTL.max.L = 10,
  xQTL.weight = NULL,
  xQTL.cred.thres = 0.95,
  estimate_residual_method = "MoM",
  unmappable_effects = "inf"
)
```

Arguments

bX
: A matrix of effect estimates from the exposure GWAS.

bXse
: A matrix of standard errors of effect estimates from the exposure GWAS.

LD
: The linkage disequilibrium (LD) matrix.

xQTL.Nvec
: The vector of sample sizes of exposures.

xQTL.max.L
: The maximum number of single effects used to estimate the xQTL effects. Defaults to 10.

xQTL.weight
: The vector of weights used in specifying the prior weights of SuSiE. Defaults to `NULL`.

xQTL.cred.thres
: The minimum empirical posterior inclusion probability (PIP) used in getting credible sets of xQTL selection. Defaults to `0.95`.

estimate_residual_method
: The method used for estimating residual variance. For the original SuSiE model, "MLE" and "MoM" estimation is equivalent, but for the infinitesimal model, "MoM" is more stable. We recommend using "Servin_Stephens" when n < 80 for improved coverage.

unmappable_effects
The method for modeling unmappable effects: "none", "inf".

Value

A list containing the results of the MRBEEX analysis using different methods:

bXest A matrix where each column represents R * beta_PLS for each exposure.

bXestse A matrix where each column contains the standard errors of R * beta_PLS for each exposure.

bXest0 A matrix where each column represents beta_PLS for each exposure.

bXestse0 A matrix where each column contains the standard errors of beta_PLS for each exposure.

xQTLfitList A list containing the xQTL selection results, which can be used in CisMRBEEX by setting xQTLfitList = xQTLfitList.

---

summary_generation *Generating simulated data for Mendelian randomization simulation*

Description

This function generates simulated data for Mendelian Randomization (MR) analysis, considering genetic effects, estimation errors, and horizontal pleiotropy. It allows for different distributions of genetic effects and pleiotropy, and accommodates both independent and correlated instrumental variables (IVs).

Usage

```
summary_generation(
  theta,
  m,
  Rbb,
  Ruv,
  Rnn,
  Nxy,
  Hxy,
  LD = "identity",
  non.zero.frac,
  UHP.frac = 0,
  CHP.frac = 0,
  UHP.var = 0.5,
  UHP.dis = "ash",
  CHP.effect = c(1, rep(0, length(theta) - 1)),
  effect.dis = "normal",
  cluster.index,
  prop = c(0.9, 0.09, 0.009, 9e-04),
  var = c(0.35, 0.4, 0.2, 0.05)
)
```

Arguments

`theta`	An (px1) vector of causal effects.
`m`	The number of instrumental variables (IVs).
`Rbb`	An (pxp) correlation matrix of genetic effects.
`Ruv`	An ((p+1)x(p+1)) correlation matrix of residuals in outcome and exposures; the outcome is the last one.
`Rnn`	An ((p+1)x(p+1)) correlation matrix of sample overlap; the outcome is the last one.
`Nxy`	An ((p+1)x1) vector of GWAS sample sizes; the sample size of the outcome is the last one.
`Hxy`	An ((p+1)x1) vector of heritabilities; the outcome is the last one.
`LD`	An (mxm) correlation matrix of the IVs or "identity" indicating independent IVs.
`non.zero.frac`	An (px1) vector with all entries in (0,1]; each entry is the probability of deltaj such that betaj=betaj'*deltaj.
`UHP.frac`	A number indicating the fraction of IVs affected by UHP.
`CHP.frac`	A number indicating the fraction of IVs affected by CHP.
`UHP.var`	A number indicating the variance attributed to UHP.
`UHP.dis`	Distribution of pleiotropy effects: "ash" (default), "normal", "uniform", "t" distribution (with degree of freedom 5).
`CHP.effect`	A vector of effects corresponding to the variables correlated with the correlated horizontal pleiotropy.
`effect.dis`	Distribution of genetic effects: "normal" (default), "uniform", "t" distribution (with degree of freedom 5).
`cluster.index`	The indices of LD block.
`prop`	When UHP.dis="ash", a numeric vector of mixing proportions (weights). These are normalized. Default: c(0.9, 0.09, 0.009, 0.0009).
`var`	When UHP.dis="ash", a numeric vector where var[i] represents the variance contribution of component i: V_i = prop[i] * sigma_i^2. Default: c(0.35, 0.4, 0.2, 0.05).

Value

A list containing simulated GWAS effect sizes for exposures (bX), their standard errors (bXse), the GWAS effect size for the outcome (by), its standard error (byse), the pleiotropy effects (pleiotropy), and the true effects.

Index

block_cutoff, 2

CisMRBEE_UV, 6
CisMRBEEX, 3
cluster_snps_and_assign, 8

errorCov, 9

filter_align, 9

GWPT, 10
GWPT_Mixture, 11

mix_normal_generator, 12
MRBEE_IMRP, 14
MRBEE_LDA, 15
MRBEE_TL, 19
MRBEE_TL_UV, 21
MRBEEX_UV, 12

Sparse_Prediction, 23
summary_generation, 24